



INTRODUCTION TO HUMAN COMPUTER INTERACTION

HCI : CHAPTER 1

NORSHADILA BINTI AHMAD BADELA
JABATAN TEKNOLOGI MAKLUMAT & KOMUNIKASI,
POLITEKNIK MERSING

LEARNING OUTCOME

- ▶ Relate the interaction between human and computer
 - ▶ IDENTIFY THE FUNDAMENTAL OF COMPONENTS OF HCI
 - ▶ DESCRIBE THE IMPORTANCE OF USER INTERFACE DESIGN
- ▶ DISCUSS HUMAN INTERACTION USAGE
 - ▶ EXPLAIN HOW COMPUTER TECHNOLOGY INFLUENCES THE NATURE OF INTERACTION AND STYLE OF THE INTERFACE
 - ▶ DESCRIBE THE VARIOUS INTERFACE STYLES
 - ▶ EXPLAIN THE ROLE OF ERGONOMICS IN INTERFACE DESIGN
 - ▶ DESCRIBE MODELS OF INTERACTION

1.1 RELATE THE INTERACTION BETWEEN HUMAN AND COMPUTER

WHAT IS HCI?



"Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them."

[ACM SIGCHI Curricula for Human-Computer Interaction]

Use and Context

U1 Social Organization and Work



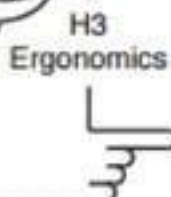
U3 Human-Machine Fit and Adaptation

U2 Application Areas

Human

H1 Human
Information
Processing

H2 Language,
Communication
and Interaction



H3
Ergonomics

Computer

C2 Dialogue
Techniques



C3 Dialogue
Genre



C4 Computer
Graphics



C5 Dialogue
Architecture



C1 Input and
Output Devices



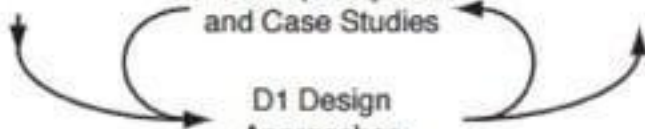
D3 Evaluation
Techniques

D4 Example Systems
and Case Studies

D1 Design
Approaches

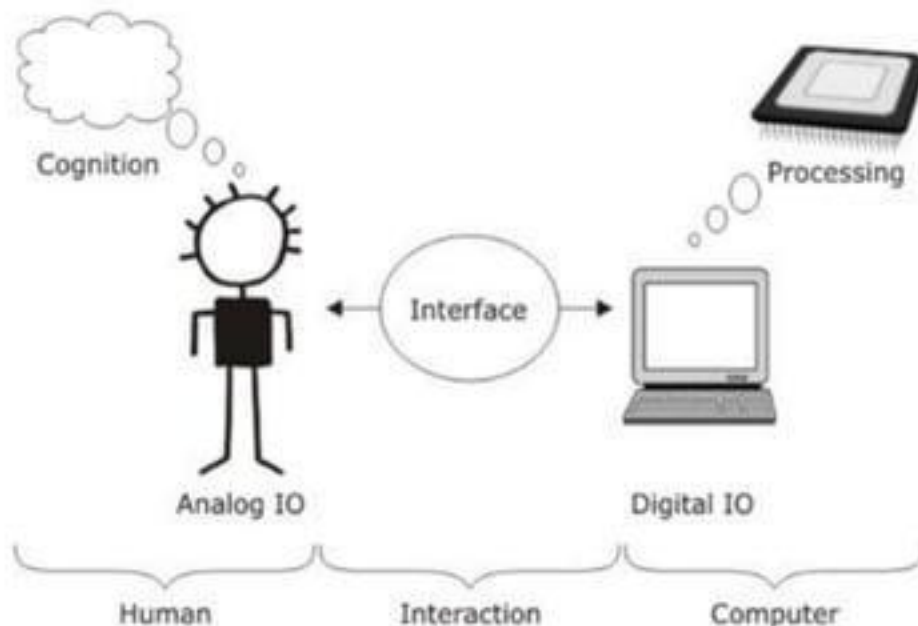
D2 Implementation
Techniques and Tools

Development Process



What is HCI?

HCI is the study of interaction between people (users) and computers



WHY IS HCI IMPORTANT?

- ▶ The study of our interface with information.
- ▶ It is not just 'how big should I make buttons' or 'how to layout menu choices'
- ▶ HCI can assist in building products/systems that are
 - ▶ *Useful*, accomplish what's required
 - ▶ *Usable*, do it easily and naturally
 - ▶ *Used*, make people want to use them
- ▶ It can affect
 - ▶ Effectiveness
 - ▶ Productivity
 - ▶ Morale
 - ▶ Safety

WHY IS HCI IMPORTANT?

- ▶ Increasing participation
 - ▶ Ensuring interfaces and systems are accessible.
- ▶ International Directives and Standards (EC Directive 90/270/EEC; ISO9241) place requirements on systems in terms of usability
- ▶ Safety and Security
- ▶ improve productivity of individuals and organizations - cost reduction, improve support, organizational enhancement
- ▶ human responses : satisfaction, no machine stress
- ▶ organization: quality and initiative, flexibility

brainstorming

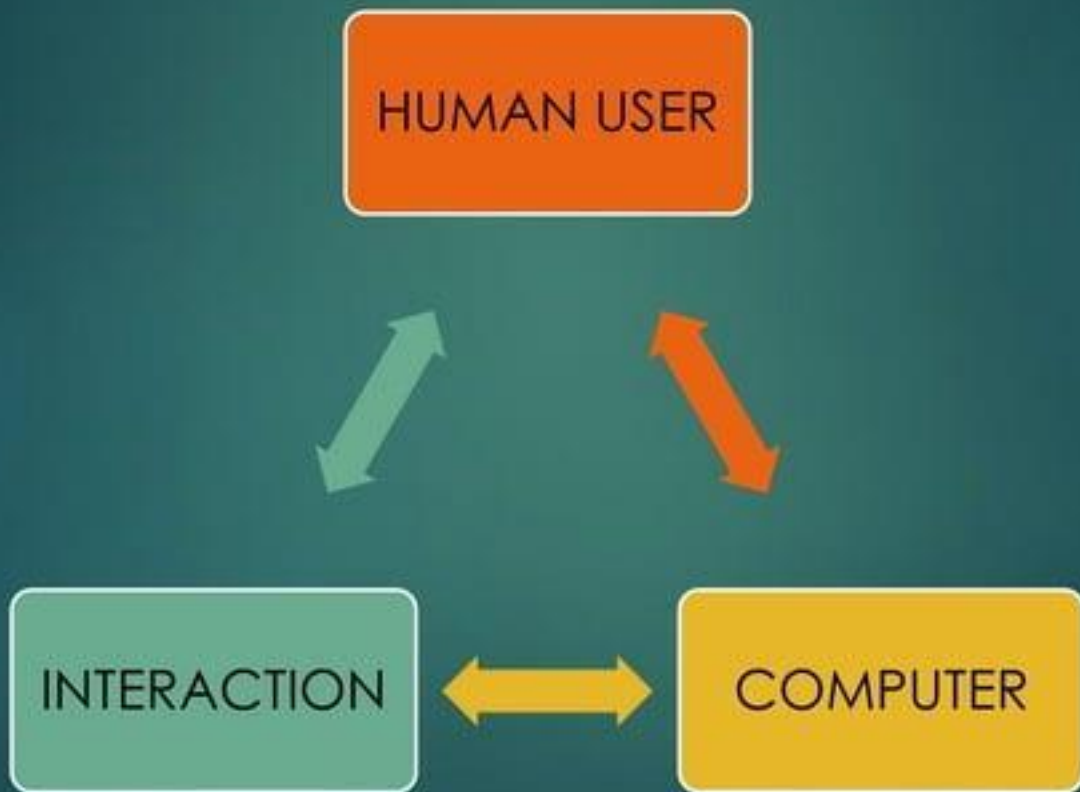
- ▶ Take 5 minutes for everyone to write down one common device with substantial HCI design choices and discuss with partner the pros and cons. How does it affect you or other users?

My Choice

- ▶ iPod by Apple Computers
- ▶ Pros:
 - ▶ portable
 - ▶ power
 - ▶ ease of use
 - ▶ # of controls
- ▶ Cons:
 - ▶ scratches easily
 - ▶ no speech for car use
 - ▶ proprietary



COMPONENTS OF HCI

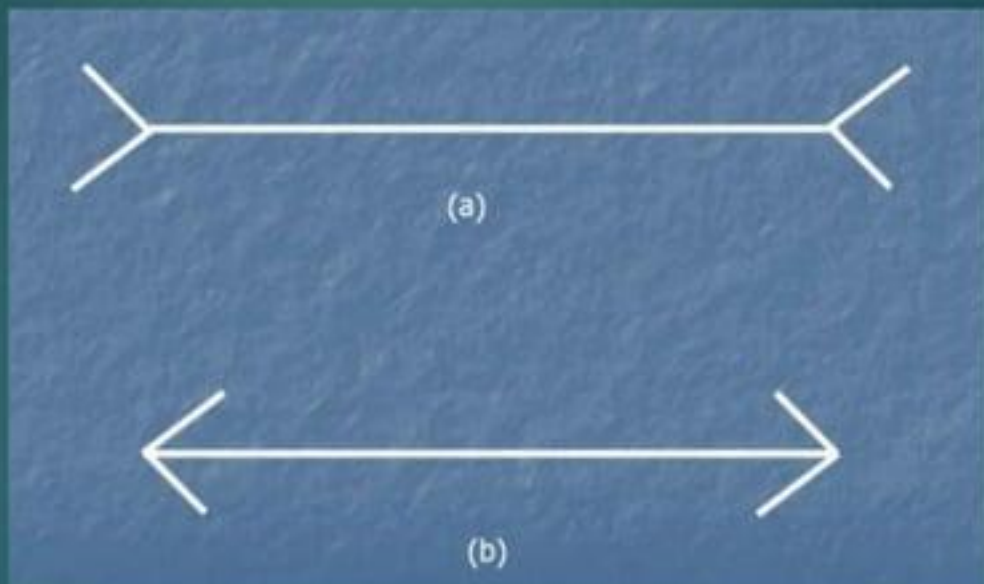


Human (User)

- ▶ Humans are limited in their capacity to process information. This has important implications for design.
- ▶ Information is received and responses given via a number of input and output channels :
 - ▶ Visual Channel - relating to seeing or sight
 - ▶ Auditory Channel - relating to the sense of hearing
 - ▶ Haptic Channel - relating to the sense of touch
 - ▶ Movement
- ▶ Information is stored in memory of :
 - ▶ Sensory memory
 - ▶ ultra-short-memory
 - ▶ 1/5 – 1/2 seconds
 - ▶ Short-term (Working) memory
 - ▶ 10 – 15 seconds – holds small amount of information; typically around 7 items or less
 - ▶ Long-term memory

brainstorming

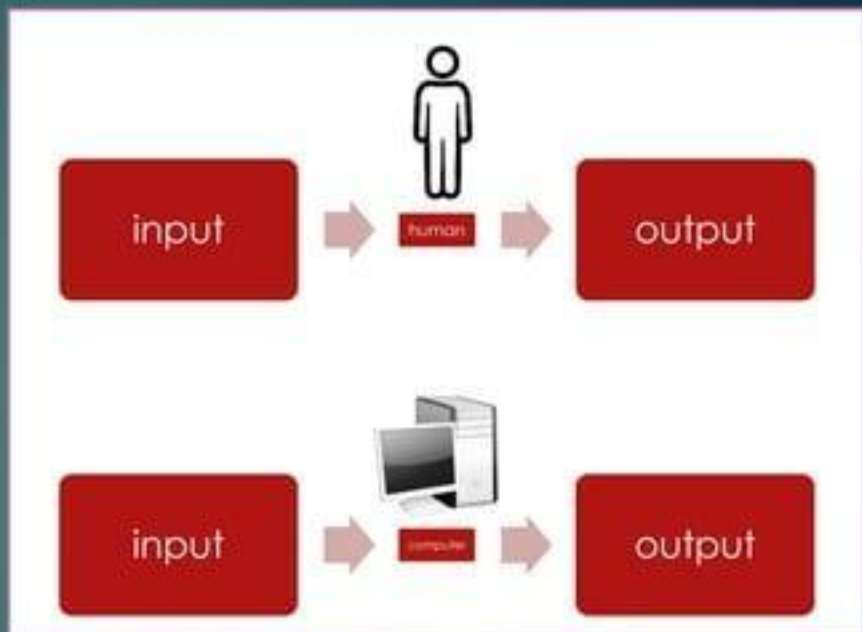
- ▶ Which line is longer?



Try to brainstorming, which one is longer? A or B? Explain your answer.

Computer

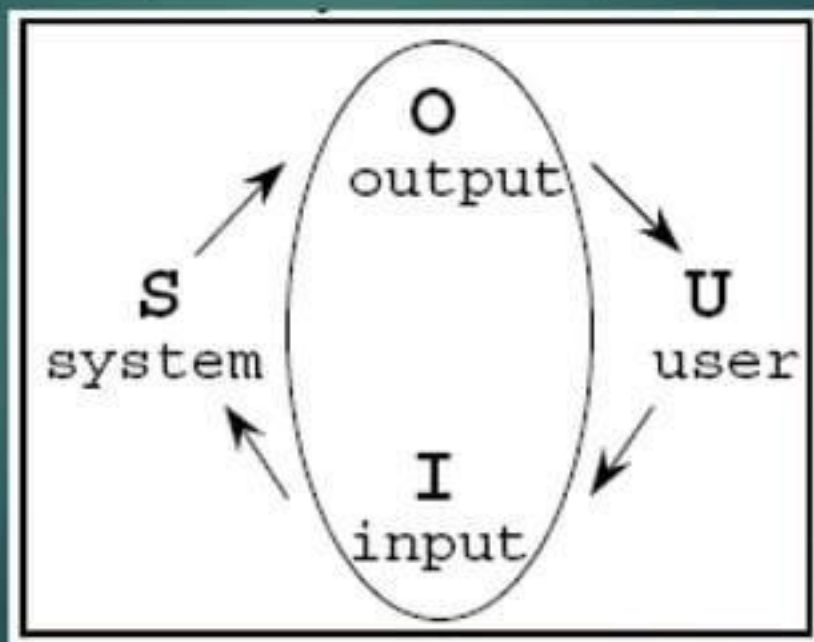
- ▶ There is not much difference in Human and Computer
- ▶ Computer consist of :
 - ▶ Input Devices
 - ▶ Output Devices
 - ▶ Memory
 - ▶ Processing
- ▶ Computer can be :
 - ▶ Mobile,
 - ▶ Spacecraft Cockpit,
 - ▶ Microwave Oven,
 - ▶ VCRs etc
- ▶ "HCI is about how to allow humans and computers to interact toward some common goal"
- ▶ Humans and computers are similar in that they both :
 - ▶ Receive input
 - ▶ Produce output
 - ▶ Process information in between



Interaction

- ▶ Communication between The User & The System
- ▶ Physical Interaction – Interaction devices
- ▶ Conceptual Interaction – Interaction styles

Interaction Framework



The Importance Of User Interface Design

- ▶ The ONLY contact medium that the user has with the system
- ▶ The interface is the system designer's way of representing the system to the user; known as conceptual model
- ▶ If the system has the confused interface – user may chose not to use the system at all OR will use it incorrectly
- ▶ A well-designed interface can increase productivity

The Model Human Processor

- ▶ Consists 3 interacting systems (each of it has its own memory & processor)

I. Perceptual processor

- ▶ Outputs into audio storage
- ▶ Outputs into visual storage

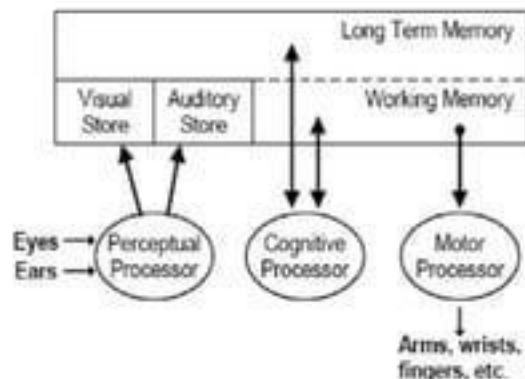
II. Cognitive processor

- ▶ Outputs into working memory
- ▶ Has access to :
 - ▶ Working memory (short-term memory)
 - ▶ Long term memory

III. Motor processor

- ▶ Carries out actions

Model Human Processor (3)



• From Brian P. Bailey, Computer Science 498bpb, Psychology of HCI
<http://www.faculty.cs.uiuc.edu/~bobailey/teaching/2004-Fall/cs498/>

Human Capabilities & Limitation

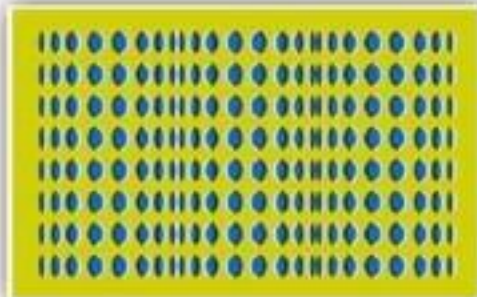
- ▶ Understanding human needs knowledge from many fields
- ▶ Processing information by human can be modelled
- ▶ Human physiology plays an important role for designing system
- ▶ Vision of human
 - ▶ Eye tracking, eyes can be tricked, pre attentive processing
 - ▶ Gestalt psychology
- ▶ Hearing
 - ▶ Audibility, pain threshold, spatial hearing
- ▶ Touch
 - ▶ Input & output
- ▶ Memory
 - ▶ Sensorial, short term (working), and long term memory
 - ▶ Short-term memory 7 ± 2 chunks
 - ▶ Long term memory : episodic and structural memory
 - ▶ Generate new information : deduction, induction, abduction

activity

Figure and ground: What do you see?



(a)



(b)



(c)

Discussion in group what do you see?
Explain your answer.

1.2 DISCUSS HUMAN INTERACTION USAGE

Interaction Devices

Input Devices

- ▶ keyboards (QWERTY etc.)
- ▶ chord keyboards, phone pads
- ▶ handwriting, speech
- ▶ Mouse, trackballs, touchpads
- ▶ Touchscreens
- ▶ Speech Input
- ▶ 3D Trackers, 3D Mouse



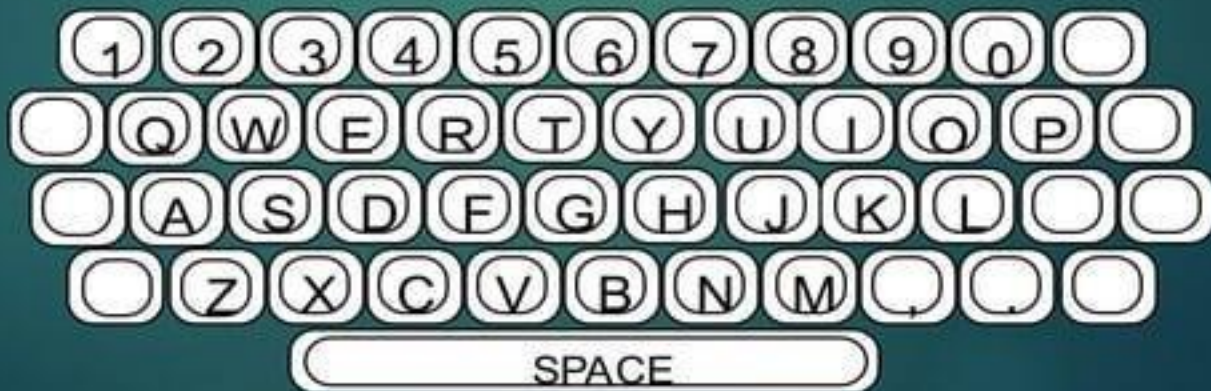
Keyboards

- Most common text input device
- Allows rapid entry of text by experienced users
- Keypress closes connection, causing a character code to be sent
- Usually connected by cable, but can be wireless



layout – QWERTY

- Standardised layout but ...
 - non-alphanumeric keys are placed differently
 - accented symbols needed for different scripts
 - minor differences between UK and USA keyboards
- QWERTY arrangement not optimal for typing
 - layout to prevent typewriters jamming!
- Alternative designs allow faster typing but large social base of QWERTY typists produces reluctance to change.



Phone Pad and T9 entry

- use numeric keys with multiple presses

2 - a b c 6 - m n o

3 - d e f 7 - p q r s

4 - g h i 8 - t u v

5 - j k l 9 - w x y z

hello = 4433555[pause]555666

surprisingly fast!



- T9 predictive entry
 - type as if single key for each letter
 - use dictionary to 'guess' the right word
 - hello = 43556 ...
 - but 26 -> menu 'am' or 'an'

Handwriting recognition

- Text can be input into the computer, using a pen and a digitizing tablet
 - natural interaction
- Technical problems:
 - capturing all useful information - stroke path, pressure, etc. in a natural manner
 - segmenting joined up writing into individual letters
 - interpreting individual letters
 - coping with different styles of handwriting
- Used in PDAs, and tablet computers ...
... leave the keyboard on the desk!



Speech Recognition

- Improving rapidly
- Most successful when:
 - single user – initial training and learns peculiarities
 - limited vocabulary systems
- Problems with
 - external noise interfering
 - imprecision of pronunciation
 - large vocabularies
 - different speakers



Numeric keypads

- for entering numbers quickly:
 - calculator, PC keyboard
- for telephones

not the same!!

ATM like phone



Output Devices

- ▶ Screens
- ▶ Printers
- ▶ Sound Output



Virtual reality and 3D visualization

- positioning in 3D space
- moving and grasping
- seeing 3D (helmets and caves)



Positioning in 3D space

▶ cockpit and virtual controls

- ▶ steering wheels, knobs and dials
... just like real!



▶ the 3D mouse

- ▶ six-degrees of movement: x, y, z + roll, pitch, yaw



▶ data glove

- ▶ fibre optics used to detect finger position



▶ VR helmets

- ▶ detect head motion and possibly eye gaze



▶ whole body tracking

- ▶ accelerometers strapped to limbs or reflective dots and video processing



3D Displays

- ▶ *desktop VR*
 - ▶ ordinary screen, mouse or keyboard control
 - ▶ perspective and motion give 3D effect
- ▶ *seeing in 3D*
 - ▶ use stereoscopic vision
 - ▶ VR helmets
 - ▶ screen plus shuttered specs, etc.



VR Headsets

- ▶ small TV screen for each eye
- ▶ slightly different angles
- ▶ 3D effect



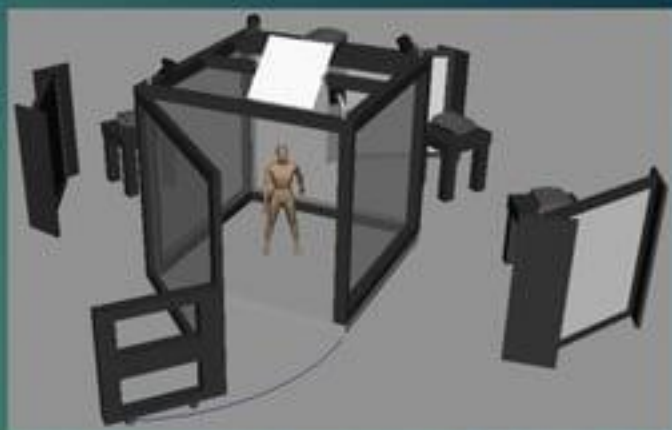
VR Motion Sickness

- ▶ *time delay*
 - ▶ move head ... lag ... display moves
 - ▶ *conflict*: head movement vs. eyes
- ▶ *depth perception*
 - ▶ headset gives different stereo distance
 - ▶ but all focused in same plane
 - ▶ *conflict*: eye angle vs. focus
- ▶ *conflicting cues* => sickness
 - ▶ helps motivate improvements in technology



Simulators and VR Caves

- ▶ scenes projected on walls
- ▶ realistic environment
- ▶ hydraulic rams!
- ▶ real controls
- ▶ other people



Physical Devices



Can be divided into :

- ▶ sound,
- ▶ touch,
- ▶ feel,
- ▶ smell
- ▶ environmental and bio-sensing

Sounds

- ▶ beeps, bongs, clonks, whistles and whirrs
- ▶ used for error indications
- ▶ confirmation of actions e.g. keyclick

Touch, feel, smell

- ▶ *touch and feeling important*
 - ▶ in games ... vibration, force feedback
 - ▶ in simulation ... feel of surgical instruments
 - ▶ called *haptic* devices
- ▶ *texture, smell, taste*
 - ▶ current technology very limited

Environment and bio-sensing

- ▶ Sensors are physical devices
 - ▶ that measure physical quantities.
 - ▶ the same property can be measured with different Sensors.
- ▶ Sensors provide raw information, which can be treated in various ways, i.e., can be processed to various levels.
- ▶ *sensors all around us*
 - ▶ car courtesy light – small switch on door
 - ▶ ultrasound detectors – security, washbasins
 - ▶ RFID security tags in shops
 - ▶ temperature, weight, location
- ▶ *... and even our own bodies ...*
 - ▶ iris scanners, body temperature, heart rate, galvanic skin response, blink rate

Example of Physical Device

- ▶ BMW iDrive
 - ▶ for controlling menus
 - ▶ feel small 'bumps' for each item
 - ▶ makes it easier to select options by feel
 - ▶ uses haptic technology from Immersion Corp.



- ▶ **Physical controls**
 - ▶ specialist controls needed ...
 - ▶ industrial controls, consumer products, etc.



Memory Capacity

- ▶ Involves re-encoding and remembering
- ▶ Human can not remember all the things - must be processed and managed
- ▶ Context is very important in memory
- ▶ Human easier to identify than recalls
 - ▶ Appearance of GUI replaces of interface based on instructions
- ▶ Human easier to remember images than words
 - ▶ Use icon rather than the words

Short-term Memory - RAM

- ▶ Random access memory (RAM)
 - ▶ on silicon chips
 - ▶ 100 nano-second access time
 - ▶ usually volatile (lose information if power turned off)
 - ▶ data transferred at around 100 Mbytes/sec
- ▶ Some *non-volatile* RAM used to store basic set-up information
- ▶ Typical desktop computers:
64 to 256 Mbytes RAM

Long-term Memory - disks

▶ *magnetic disks*

- ▶ floppy disks store around 1.4 Mbytes
- ▶ hard disks typically 40 Gbytes to 100s of Gbytes
access time ~10ms, transfer rate 100kbytes/s

▶ *optical disks*

- ▶ use lasers to read and sometimes write
- ▶ more robust than magnetic media
- ▶ CD-ROM
 - same technology as home audio, ~ 600 Mbytes
- ▶ DVD - for AV applications, or very large files

Blurring boundaries

- ▶ *PDA's*

- ▶ often use RAM for their main memory

- ▶ *Flash-Memory*

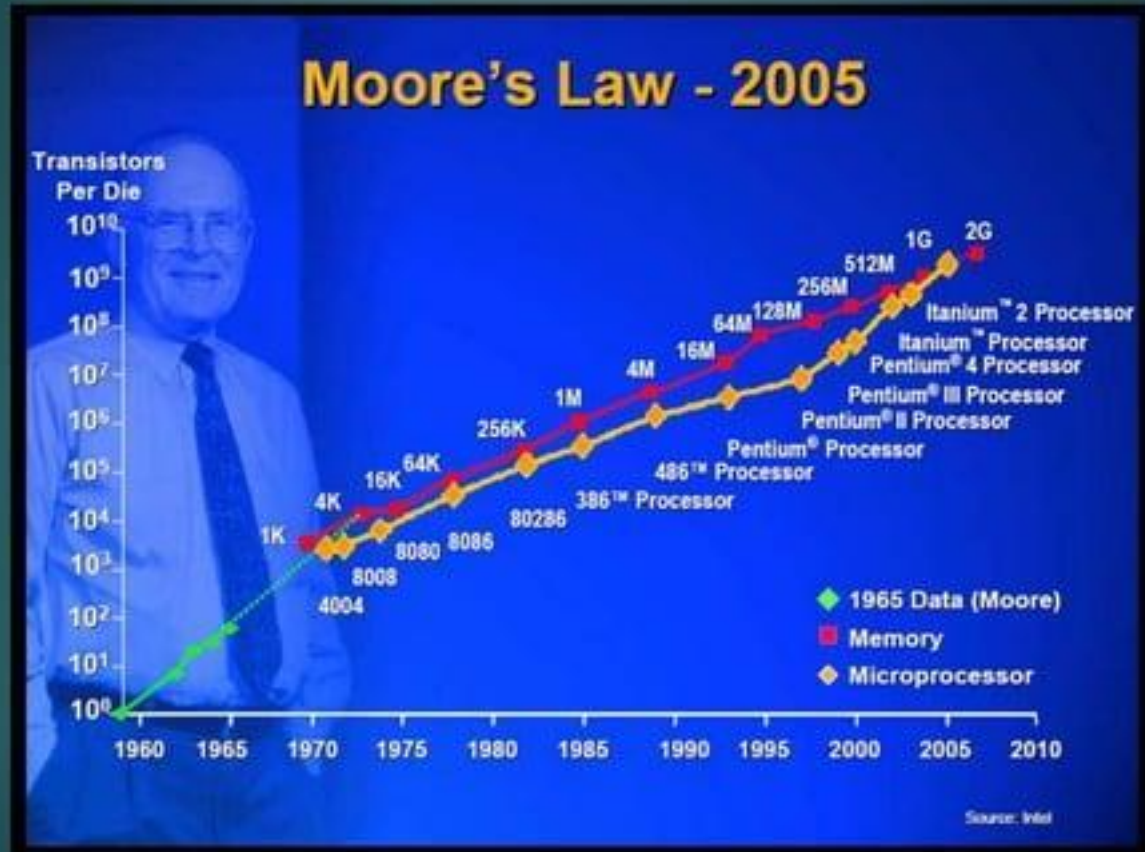
- ▶ used in PDA's, cameras etc.
 - ▶ silicon based but persistent
 - ▶ plug-in USB devices for data transfer

Finite Processing Speed

- ▶ Designers tend to assume fast processors, and make interfaces more and more complicated
- ▶ But problems occur, because processing cannot keep up with all the tasks it needs to do
 - ▶ cursor overshooting because system has buffered keypresses
 - ▶ icon wars - user clicks on icon, nothing happens, clicks on another, then system responds and windows fly everywhere
- ▶ Also problems if system is too fast - e.g. help screens may scroll through text much too rapidly to be read

Processing Speed

- ▶ finite speed (but also Moore's law)



Moore's law

- ▶ computers get faster and faster!
- ▶ 1965 ...
 - ▶ Gordon Moore, co-founder of Intel, noticed a pattern
 - ▶ processor speed doubles every 18 months
 - ▶ PC ... 1987: 1.5 Mhz, 2002: 1.5 GHz
- ▶ similar pattern for memory
 - ▶ but doubles every 12 months!!
 - ▶ hard disk ... 1991: 20Mbyte : 2002: 30 Gbyte
- ▶ baby born today
 - ▶ record all sound and vision
 - ▶ by 70 all life's memories stored in a grain of dust!

The various interface style

- a. Command line interface
- b. Menus
- c. Natural language
- d. WIMP interface: Windows, icon, menus and pointers.
- e. Question/answer and query dialog
- f. Form-fills and spreadsheet
- g. Point-and-click interfaces
- h. 3D interfaces
- i. Web navigation

a. Command Line Interface

- operating systems - DOS/UNIX
- short instructions for efficiency/repetition
 - ✓ good for experts
 - ✗ poor for novices
- hard to remember, so choose command names carefully
- includes keyboard shortcuts and function keys.
- simple, quick
- developed from Teletypewriters (TTY)
- not for complex tasks
- useful for operating system tools, compilers, ...
- relies on recall rather than recognition

<u>Operation</u>	<u>DOS command</u>	<u>Unix program</u>
directory list	dir	ls
display file	type	cat
rename file	rename	mv
search in file	findstr	grep

b. Menus

- a set of options
 - ✓ no need to remember, only recognize) options must be self-explanatory
- they can take up space so we have
 - pull-down/pop-up/pin-up/cascading/pie
- Selection by:
 - ▶ numbers, letters, arrow keys, mouse
 - ▶ – combination (e.g. mouse plus accelerators)
- Often options hierarchically grouped
 - ▶ – sensible grouping is needed
- Restricted form of full WIMP system

c. *Natural Language*

- ▶ **Method** whereby inputs to and outputs from a computer-based application are in a conventional spoken language such as English.
- ▶ Current implementations are tedious and difficult to work with, not as viable as other interaction methods.
- ▶ Applications for Natural Language
 - ▶ Speech Input
 - ▶ Hands-free operation
 - ▶ Poor Lighting Situations
 - ▶ Mobile Applications
 - ▶ In the home
 - ▶ Patients and disabled
 - ▶ Speech Output
 - ▶ On-board navigational systems
- ▶ Two areas of development
 - ▶ Speech recognition
 - ▶ Semantics
 - ▶ Grammar issues
 - ▶ Vague meanings
 - ▶ Contradictory statements

d. *WIMP Interface*

- ▶ Default interaction style for majority of interaction computer systems, especially PCs and desktop machines.
- ▶ WIMP (Windows, Icons, Menus, Pointers) systems replace typed commands.
- ▶ They have the following features:
 - ▶ objects are visible and directly manipulable
 - ▶ rapid, reversible, incremental actions

- 
- Why are these systems popular?
 - ❖ novices learn quickly
 - ❖ programs tend to have the same look and feel
 - ❖ based on recognition not remembering
 - ❖ provide immediate feedback
 - ❖ contain recognizable widgets, e.g. buttons, dialogs
 - ❖ provide a feeling of control
 - There are conceptual problems; e.g. the gulfs of execution and evaluation.

Windows

- ▶ Independent terminals in their own right.
- ▶ Contain text/graphics and can be moved or resized
- ▶ More than one window can be on screen at once



- 
- ▶ Windows may also be tiled, when they adjoin but do not overlap each other.
 - ▶ Windows have various things associated with them that increase their usefulness
 - ▶ Ex:
 - ▶ Scrollbars –allowing the user to move the contents of the window up and down/from side to side
 - ▶ Title bar – identifying it to the user
 - ▶ Special box in the corners – to aid resizing, closing or making as large as possible.

Icons

- ▶ a graphical representation of an object
- ▶ can be dragged and dropped
- ▶ Icons embody the idea that different people have different cognitive styles.
- ▶ Some users prefer text-oriented views and some prefer graphics.
- ▶ An icon is an image, picture or symbol representing a concept.
- ▶ Can take many forms
 - ▶ Realistic
 - ▶ Highly stylized
 - ▶ Arbitrary symbol(difficult for user to interpret)



- **Consider these guidelines when creating or allocating icons:**

- ❖ *represent the object or action in a familiar manner*
- ❖ *limit the number of icons*
- ❖ *make the icon stand out*
- ❖ *try 3D icons*
- ❖ *make each one clearly visible from the background*
- ❖ *ensure icon 'family' harmony*
- ❖ *group icons appropriately*
- ❖ *add information to show use, e.g. tooltips*

Menus

- ▶ a list of command buttons
- ▶ pull-down or pop-up
- ▶ Presents a choice of operations or services that can be performed by the system at a given time
- ▶ The name used for the commands in the menu should be meaningful and informative.
- ▶ Main Menu can be visible to the user
- ▶ Website use variety of menu bar locations (top, bottom and either side of the scene)
- ▶ Main menu can be hidden and upon request it will pop up onto the screen

Types of menu:

- ▶ **Pop up menu** – allow one to examine properties of particular on screen objects
- ▶ **Pull down menu** – dragged down from the title at the top of the screen, moving the pointer into the title bar area and pressing the button.
- ▶ **Fall down menu** – menu automatically appears when the mouse pointer enters the title bar without pressing the button
- ▶ **Pin up menu** – “pinned” to the screen

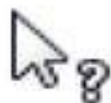
Example of menu:



Pull down menu

Pointers

- ▶ mouse or similar pointing device
- ▶ Example: Text pointer and Mouse pointer
- ▶ Most important component to point and selecting things such as icons.
- ▶ Different shapes of cursor are often used to distinguish modes
- ▶ ex:
 - ▶ normal cursor (arrow) – change to cross hair when drawing a line
 - ▶ Watch/hourglass cursor – system busy reading a file



e. Question/Answer & Query

- ❖ A simple mechanism for providing input to an application in a specific domain
- ❖ input is constrained - ATM, "wizard" dialogue
- ❖ good for novices but restricted functionality
- ❖ next question/action depends on last answer



Differences between Question/Answer & Query Dialog Box

Question/ Answer	Query Dialog
A simple mechanism for providing input to an application in as specific domain	Construct queries to retrieve information from a database
User is led through interaction step by step via series of questions	Require understanding of database structure and language syntax
Limited in functionality and power	Often do not provide direct conformation of what was requested, so the only validation the user has is the result of the search
Appropriated for restricted domains and for novice or casual users	Effective use require some expertise

f. Form-Fills/Spreadsheets

- ▶ Primarily for data entry but can also be useful in data retrieval applications
- ▶ hard-copy metaphor
- ▶ used for data entry & specifying a data retrieval operation
- ▶ good for novices as it is familiar (similar to paper form)
- ▶ need to allow editing of errors
- ▶ need to allow easy movement between fields

Microsoft Excel - Green Surface.xls							PEAKS
File Edit Format Tools Project Window Help							100%
Production Totals							
	1	2	3	4	5	6	
1	Production Totals						
2		Item	Place Count	Volume	Length	% Vol	% Length
3		Wired	1,061	8862.5	10871.0	78.0%	100.0%
4		Corded	122.0	883.9	16384.0	90.0%	96.6%
5		Wt Sock Pender	1235	3540.3	6795.0	3.7%	3.6%
6		Wt Box Pender	8	69.3	112.0	0.1%	0.1%
7		Chains	25	30.4	162.0	0.0%	0.1%
8		Total Outfeed	13566	89989.5	109994.0	95.8%	90.3%
9		Total Loss	-25	8803.9	16223.0	0.2%	9.7%
10		Manual Slew	0				
11		Logit Fun	20780				
12		Time Running	7:30:45				
13		PLC Time Shipped	0:15:30				
14		Operator Time Shipped	3:52:35				
15		Tally Factor Used	87.0%				
16							
Ready							Quind

Registration Form

Registration

ID:
 Name:

Personal Contact Details

Email: Phone:

Security of Travel

Date Applied:
 Status Type:
 Status Type (for sub-voting): Next to:
 Passport No.:
 Expiry Date:

Health Status and Symptom

Sub-symptom:
 Health By:
 Symptom By:

g. *Point and click interfaces*

- ▶ Closely related to the WIMP style
- ▶ More closely tied to ideas of **hypertext**
- ▶ not tied to mouse-based interfaces – extensively used in touchscreen information system
- ▶ Popularized by world wide web pages
- ▶ Incorporate all the above types of point and click navigation:
 - ▶ Highlighted words,
 - ▶ Maps
 - ▶ Iconic buttons

h. 3D Interface



- 
- ▶ human-computer interaction in which the user's tasks are performed directly in a 3D spatial context.
 - ▶ VR is only part of a range of 3D techniques available to the interface designer.
 - ▶ Simple technique for WIMP elements, buttons, scroll bars, etc are given a 3D appearance using shading
 - ▶ Complex technique uses interfaces with 3D workspace
 - ▶ Object in perspective when at an angle to the viewer and shrink when they 'further away'

i. Web navigation

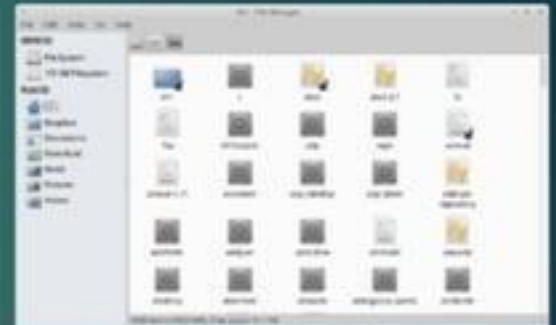
- ▶ Two basic interaction styles
 - ▶ Link-based navigation
 - ▶ Sensitive to articulatory distance
 - ▶ Ambiguous link labels increase the gulf of evaluation
 - ▶ Search
 - ▶ Sensitive to semantic distance
 - ▶ Inadequate search engine algorithms increase the gulf of execution
 - ▶ Slight advantage in development of mental models
- ▶ Readers need a sense of context of their place within an organization of information.

Styles of web navigations

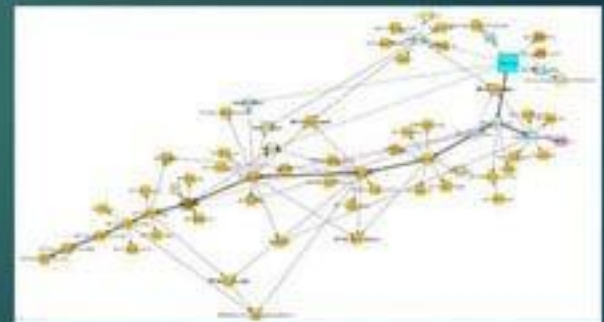
a. **Text links**– The anchor text, link label, link text, or link title is the visible, clickable text in a hyperlink



b. **Navigation bar** – A navigation bar or (navigation system) is a section of a website or online page intended to aide visitors in travelling through the online document

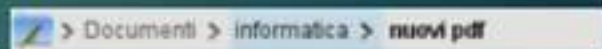


c. **Sitemap** – A site map (or sitemap) is a list of pages of a web site accessible to crawlers or users. It can be either a document in any form used as a planning tool for Web design, or a Web page that lists the pages on a Web site, typically organized in hierarchical fashion



Styles of web navigations

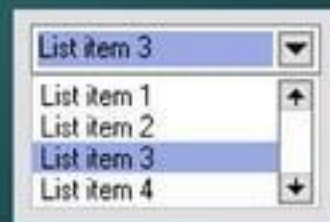
d. **Breadcrumbs** - Breadcrumbs or breadcrumb trail is a navigation aid used in user interfaces. It allows users to keep track of their locations within programs or documents. The term comes from the trail of breadcrumbs left by Hansel and Gretel in the popular fairy tale



e. **Named anchor** - An anchor element is called an anchor because web designers can use it to anchor a URL to some text on a web page. When users view the web page in a browser, they can click the text to activate the link and visit the page whose URL is in the link.



d. **Dropdown Menu**: In computing with graphical user interfaces, a **dropdown menu** or **drop-down menu** or **drop-down list** is a user interface control GUI element ("widget" or "control"), similar to a list box, which allows the user to choose one value from a list.



Human factors in interface design

- ▶ **Limited short-term memory**

- ▶ People can instantaneously remember about 7 items of information. If you present more than this, they are more liable to make mistakes.

- ▶ **People make mistakes**

- ▶ When people make mistakes and systems go wrong, inappropriate alarms and messages can increase stress and hence the likelihood of more mistakes.

- ▶ **People are different**

- ▶ People have a wide range of physical capabilities. Designers should not just design for their own capabilities.

- ▶ **People have different interaction preferences**

- ▶ Some like pictures, some like text.

Ergonomics in interface design

Ergonomics (Human Factors)

- Study of the physical characteristics of interaction
- Also known as human factors – but this can also be used to mean much of HCI!
- Ergonomics good at defining standards and guidelines for constraining the way we design certain aspects of systems

Role of Ergonomic in Interface Design

- This is a huge and established field; we will consider briefly some aspects:
 - ❑ arrangement of controls
 - ❑ e.g. controls grouped according to function or frequency of use, or sequentially
 - ❑ physical environment
 - ❑ e.g. seating arrangements adaptable to cope with all sizes of user
 - ❑ health issues
 - ❑ e.g. physical position, environmental conditions
 - ❑ (temperature, humidity), lighting, noise
 - ❑ use of colour
 - ❑ e.g. use of red for warning, green for okay, awareness of colour-blindness, etc

Ergonomics: the arrangement of controls

- Control layout is important
 - ❑ Safety critical systems: poor layout) disaster!
 - ❑ Routine applications: poor layout) inefficiency, user dissatisfaction, poor mental model building etc..
- Controls can be and laid out in various ways:
 - ❑ functional - task related controls grouped together
 - ❑ sequential - layout in order of use
 - ❑ frequency - common controls easy to access
- Other factors
 - ❑ Controls should be easy to reach
 - ❑ Controls should not be so close to each other that they hamper usage
 - ❑ 'Dangerous' controls should be hard to reach - prevents accidents

Ergonomics: The physical environment

- Unsatisfactory working conditions can at best lead to stress and dissatisfaction
- 'Physical' here means the kinds of things physicists know and love - heat, light, noise, dusts, chemicals, and so on.
- For example, there's a thermal comfort range which suits people best - unless they are doing hard physical work, in which case they might prefer a cooler range. Similarly with noise; at night, you might want things quiet so that you can sleep; in a club, you might want it a bit louder!
- About understanding the effects of these aspects of the environment on people, and in particular, the harmful effects.

Ergonomics: Health issues

- and at worst harm workers' health. Some factors to consider:
 - ❑ **physical position** - should be comfortable
 - ❑ **temperature** - should not be extreme
 - ❑ **Lighting** - should be low-glare & sufficient
 - ❑ **Noise** - should not be excessive; high levels hamper perception
 - ❑ **Time** - don't expect extended use of an interactive system

Ergonomics: Colour

- Colour is a powerful cue, but it is easy to misuse.
- It should not be applied just because it is available.
- Topics:
 - Colour Vision & Perception
 - Principles & Guidelines

Colour Vision

- the eye consists of millions of photo receptors sensitive to light
- two types of photo receptors
 1. rods
 - ❑ not sensitive to colour
 - ❑ high density at periphery
 - ❑ highly sensitive
 - ❑ low resolution

Colour Vision

2. cones

- ❑ sensitive to colour; different cones for red, green and blue light
- ❑ high density in centre (fovea)
- ❑ less sensitive - can tolerate bright light

How are colours generated?



- *Subtractive colour system*
 - ❖ Non-luminous objects (e.g. paper) selectively absorb and reflect different wavelengths of light, creating the perception of colour.
- *Additive colour system*
 - ❖ Luminous objects (e.g. CRT screen) generate colour by addition of Red/Green/Blue.

Other Colour models

- As well as the RGB system, a number of other descriptive models are in use.
 - ❑ The most common other model is the HLS (or HSB) system
 - ❑ HLS describes more closely the colours that we actually can see. (Many colours that we can see are not describable in the RGB system.)

Other Colour models

- The HLS colour model has three (3) dimensions:
 1. hue - the basic component
 2. saturation - the degree to which the hue differs from a neutral gray
 3. lightness indicates the level of illumination:

Colour Principles & Guidelines

- have some other redundant cue
 - ❑ optimal combinations are known
 - ❑ include a bright colour in the foreground
 - ❑ best background - black
 - ❑ worst background - brown or green
 - ❑ use colour sparingly, design in B&W
 - ❑ use colour to group/highlight information
 - ❑ use colour to support search tasks
 - ❑ avoid using colour in non-task-related ways

Colour Principles & Guidelines

- allow customisation
 - ❑ ensure colours differ in lightness (aids colourblind users)
 - ❑ limit colour to eight (8) distinct colours; four (4) preferred
 - ❑ avoid saturated blues for text
 - ❑ choose foreground and background colour with care
 - ❑ colours are hard to distinguish when objects are small, far apart, or close
 - ❑ on colour spectrum

Model Of Interaction

- Why develop a model for interaction?
 - ✓ To help us to understand an interactive dialogue.
 - ✓ To identify likely difficulties.
 - ✓ To provide a framework to compare different interaction styles.

Some concepts:

- Users want to achieve goals in some domain.
- Operations in the domain are tasks.
- Task analysis investigates the problem in terms of domain, goals, intentions, tasks
- The system and the user have different languages
 - The core language describes computation aspects of the domain
 - The task language describes psychological aspects of domain

Mental Model

- ▶ What is a mental model?

- Norman (1988, p. 17):

... the models people have of themselves, others, the environment, and the things with which they interact.

- ▶ Mental models are used to explain *observable events in terms of unobservable structures and events*

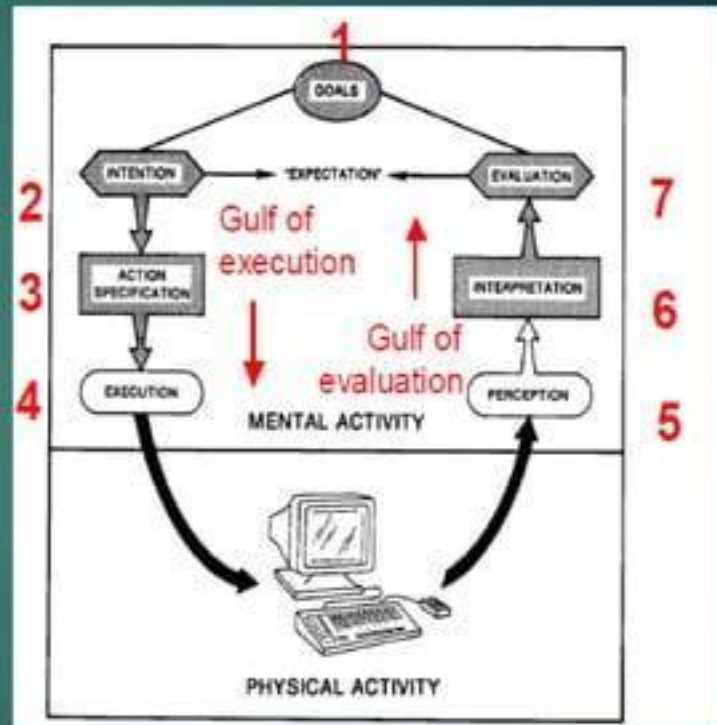
Formation of Mental Model

Cognitive model: Theory of action Norman's (1986) 7 stage model of activity for users at an interface

- a) *establishing the goal - task language; imprecise*
- b) *forming the intention - specific*
- c) *specifying the action sequence*
- d) *executing the action*
- e) *perceiving the system state*
- f) *interpreting the system state*
- g) *evaluating the system state with respect to the goals and intentions*

An example: reading breaking news on the web

1. Set goal: "find out about breaking news"
 - decide on news website
2. Form an intention
 - check out a news website
3. Specify what to do
 - search with Google for Fairfax News website
4. Execute action sequence
 - find and select suitable link
5. Check what happens at the interface
 - see a new page pop up on the screen
6. Interpret it
 - confirm that the web page is correct one
7. Evaluate it with respect to the goal
 - read breaking news



Interface Problems:

- Since the human and computer do not recognize the same concepts (speak the same language) interfaces cause problems. These problems can be described in terms of:
 - ❖ *gulf of execution* - difference between user determined action formulation and the actions allowed by system
 - ❖ *gulf of evaluation* - difference between physical presentation of system state and user expectation

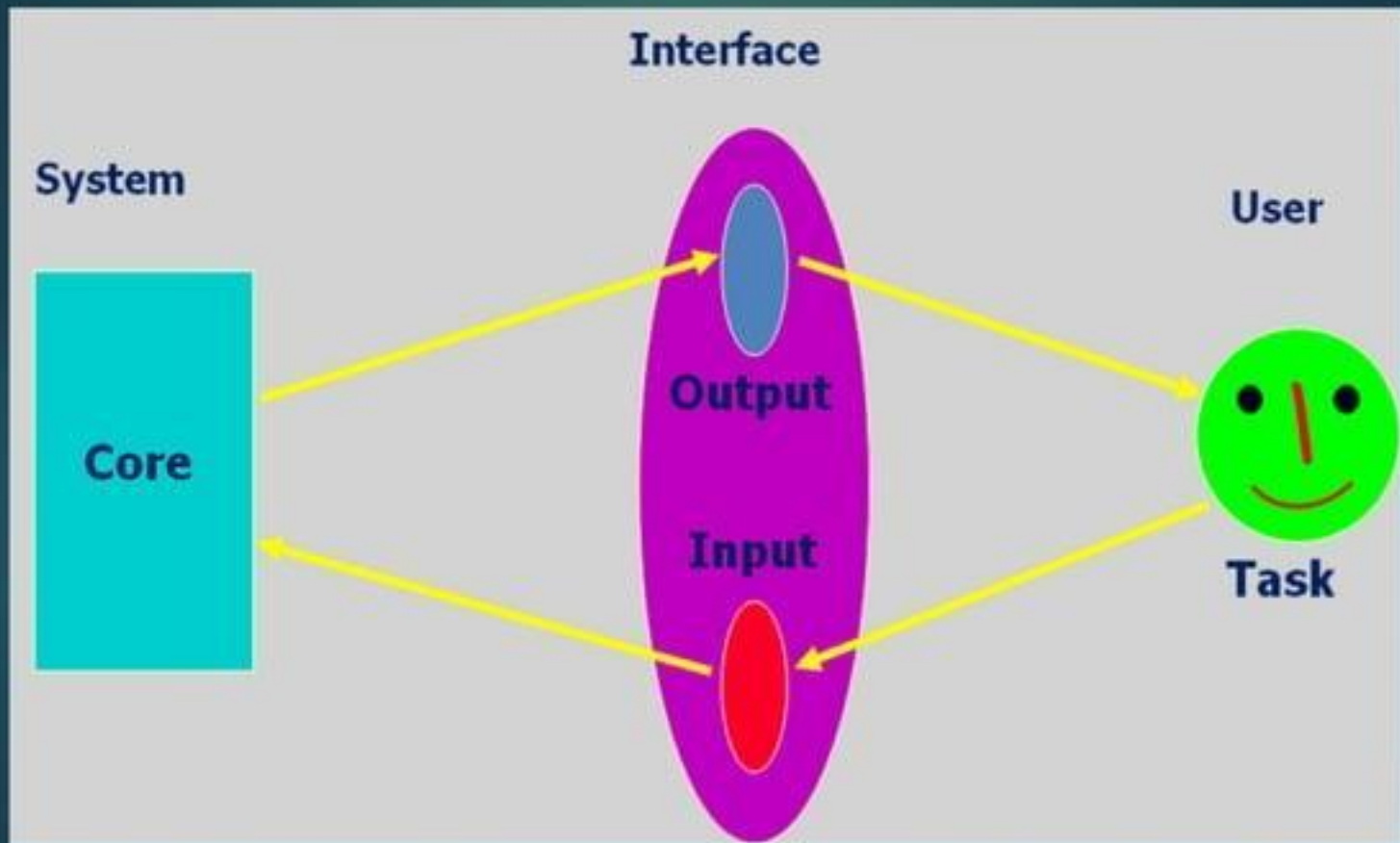
► These gulfs can be `bridged':

- users can change to suit the interface
- designers can design "knowing the user"
- users can change their interpretation of system responses
- designers can change output characteristics

The Interaction Framework

- ▶ components
 - ▶ system, user, input, output
 - ▶ input and output together form the interface
 - ▶ each of the components may have its own language to describe the objects and actions it is concerned with

Interaction Framework



- 
- ▶ They considered four (4) components, each with its own language.
 - ▶ This more accurately models the overall interaction and leads to situation
 - ▶ analyses that locate sources of interface problems

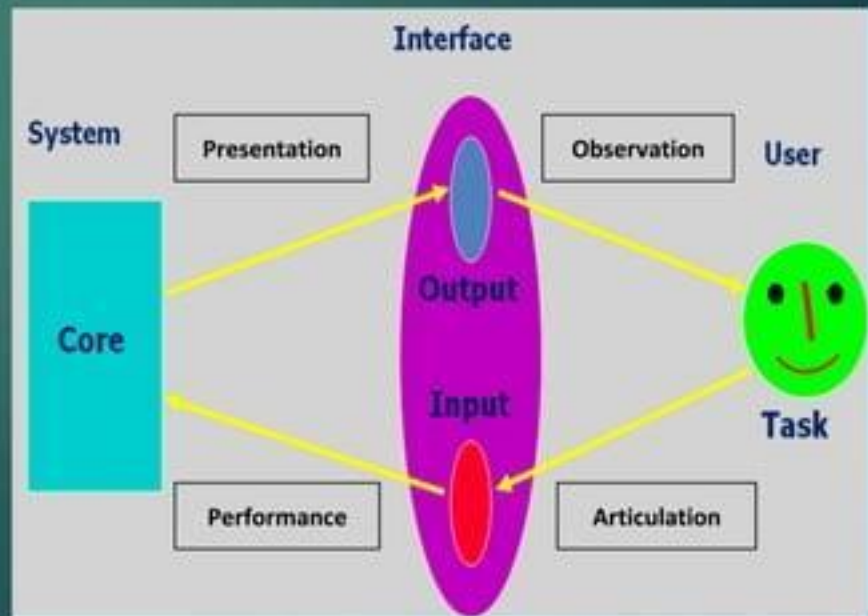
- 
- Interaction problems are explained as language translation difficulties
 - User - Input: (articulating a goal)
 - ❑ How easy is it to translate a goal requirement into the input language? e.g. Difficult: bank of light switches, stovetop element controls
 - ❑ Easy: virtual reality system
 - Input -System
 - ❑ Can all system stimuli be articulated by user language?
 - ❑ Consider remote control (or front panel) with limited functions.

- 
- System - Output (execution & evaluation)
 - ❑ Can system output device provide a complete view of system state? e.g.
 - ❑ Consider document editing with limited view of data
 - Output - User (interpretation by user)
 - Is information presented to user in a way that is easy to interpret. e.g.
 - ❑ Difficult to read unmarked analog clock.
 - ❑ Difficult to observe result of hierarchical system like copying using command line interface

Interactive Cycle

Translations Between Components

- articulation
user translates task intentions into the input language
- performance
input language is translated into stimuli for the system
- presentation
system activities are translated into output language
- observation
output language is translated into the user's task model



Example: Light in a Room

- ▶ controlling the lighting in a room
 - ▶ articulation: "I'm going to bed now, so I better turn off the light in the living room. To do this, I need to flip the switch."
 - ▶ task language: turn lights on/off
 - ▶ input language: flip switch
 - ▶ system language: close/open circuit for light bulbs
 - ▶ output language: lights on/off
- ▶ translations
 - ▶ articulation
 - ▶ user decides to turn on the light, and flips a switch
 - ▶ performance
 - ▶ flipped switch closes the circuit
 - ▶ presentation
 - ▶ light bulb emits light
 - ▶ observation
 - ▶ user notices that the light is on
- ▶ frequent problem
 - ▶ multiple switches in large rooms

- 
- ▶ Interactivity is the defining feature of an interactive system, e.g. the interface semantics and closeness to real-time interaction (speech recognition, visualisation, menu dynamics).
 - ▶ In older systems, order of interaction is preemptive. Newer systems still have some of these features, e.g. modal forms.
 - ▶ Of course all interaction occurs in some wider social and organizational context.

- 
- ▶ People are usually involved and there are issues of desire to impress, competition and fear of failure.
 - ▶ Motivation will reduce if systems do not match requirements but new technology may increase motivation if systems are well designed and integrated with the user's work.

Comments:

- ▶ These two frameworks are quite general.
- ▶ Frameworks help us to understand the interaction process.
- ▶ Frameworks help us to judge overall usability of an entire interactive system.
- ▶ Evaluation of a system can only be with respect to a set of goals and tasks
- ▶ which the system is being used to achieve. (E.g. Word vs LATEX)